RAFFLES INSTITUTION
RAFFLES PROGRAMME 2026
YEAR 3 MATHEMATICS 2
TEST 2 MOCK PRACTICE PAPER 2

20

Parent's Signature | Duration: 30 minutes

Name: ______________________________ ( ) Class: 3 ( ) Date: __________

Instructions:

1. Answer **all** questions in the space provided.
2. This paper consists of printed pages.
3. The use of a calculator is allowed unless otherwise stated.
4. Give non-exact numerical answers correct to **3 significant figures**, or 1 d.p. for angles, unless stated otherwise.
5. Omission of essential working will result in loss of marks.

1. Solve the equation

$$\log_4(3x) - \log_{16}(3x - 1) = \tfrac{1}{2}.$$

[4]

Working space

2. Solve the equation

$$2e^{2x+1} = e^{x+1} + 15e,$$

leaving your answer in exact form.

[3]

Working space

3.

(i) Sketch the graph of $y = \log_{0.5}(x + 3)$, indicating clearly all axial intercepts and the asymptote. **[3]**

Sketch here

(ii) By drawing a suitable straight line on your sketch in **(i)**, determine the number of solutions to the equation **[3]**

$$log_{0.5}(x + 3) = -3x - 5.$$

Working space

4. It is given that $\log_8 x = p$ and $\log_8 y = q$.

(i) Find $\log_8(x^3y^{-2})$ in terms of p and q. **[1]**

(ii) Find $\log_4(x/y)$ in terms of p and q. **[3]**

Working space

(iii) Find m in terms of p if $64^{2m} = x$. **[2]**

Working space

(iv) Given also that $\log_8 z = r$, express $\log_x z$ in terms of p and r. **[2]**

Working space

~ End of Paper ~

WORKED SOLUTIONS — MOCK PAPER 2

Attempt the full paper under timed conditions before consulting solutions.

Question 1 [4 marks]

M1 Since $\log_4 16=2$: $\log_{16}(3x-1)=\log_4(3x-1)/2$. Equation: $\log_4(3x)-\log_4(3x-1)/2=1/2$. Multiply by 2: $2\log_4(3x)-\log_4(3x-1)=1$.

M1 Condense: $\log_4[(3x)^2/(3x-1)]=1 \Rightarrow 9x^2/(3x-1)=4$.

M1 $9x^2=12x-4 \Rightarrow 9x^2-12x+4=0 \Rightarrow (3x-2)^2=0$.

A1 $\boldsymbol{x=2/3}$. Check: $3(2/3)=2>0$ ✔, $3(2/3)-1=1>0$ ✔.

Why this works: The conversion $\log_{16} \rightarrow \log_4$ uses $\log_4 16=2$, giving a factor of 1/2. Multiplying through by 2 clears the fraction. The resulting quadratic is a perfect square, giving a unique solution.
Common error: Reversing the factor: $\log_{16}x = 2\log_4 x$ is WRONG. Since $16=4^2$, $\log_{16}x = \log_4 x/\log_4 16 = \log_4 x/2$.

Question 2 [3 marks]

M1 Divide both sides by e: $2e^{2x}=e^x+15$. Let $u=e^x$: $2u^2-u-15=0$.

M1 $(u-3)(2u+5)=0 \Rightarrow u=3$ (accept) or $u=-5/2$ (reject, $e^x>0$).

A1 $e^x=3 \Rightarrow \boldsymbol{x=\ln 3}$.

Why this works: Dividing by e first is the simplest approach: it turns e^{2x+1} into $e^{2x}\cdot e/\cdot e=e^{2x}$. The substitution $u=e^x$ then gives a standard quadratic.

Question 3 [6 marks]

(i) Base $0.5<1 \Rightarrow$ decreasing. Asymptote: $\boldsymbol{x=-3}$ (dashed). x-intercept: $x+3=1 \Rightarrow \boldsymbol{x=-2}$. y-intercept: $\log_{0.5}3=-\log_2 3\approx-1.58$.

M1 **(ii)** The required line is $\boldsymbol{y=-3x-5}$. This is the right-hand side of the equation written as a straight line.

M1A1 Draw $y=-3x-5$ on the axes. Slope -3, y-intercept -5, x-intercept $-5/3$. It intersects the decreasing log curve once: **1 solution**.

Question 4 [8 marks]

B1 **(i)** $\log_8(x^3y^{-2}) = 3p - 2q$.

M1 **(ii)** $\log_8 4=\log_8 4$. Since $8=2^3$ and $4=2^2$: $\log_8 4=2/3$. So $\log_4 x=p\div(2/3)=3p/2$. Similarly $\log_4 y=3q/2$.

A1 $\log_4(x/y)=3p/2-3q/2=\boldsymbol{3(p-q)/2}$.

M1A1 **(iii)** $64^{2m}=x \Rightarrow (8^2)^{2m}=8^p \Rightarrow 8^{4m}=8^p \Rightarrow 4m=p \Rightarrow \boldsymbol{m=p/4}$.

M1A1 **(iv)** $\log_x z=\log_8 z/\log_8 x=\boldsymbol{r/p}$.

Why this works: In (iv), change-of-base with base 8 converts $\log_x z$ to known quantities. This works even when the base x is itself unknown, as long as we know $\log_8 x=p$.